Improved Current-Source Circuits

Two performance parameters of the BJT current source need improvement.

↳ The first is the dependence of I_0 on β, which is a result of the error in the mirror current-gain introduced by finite BJT β.

** Effect of finite β becomes more severe as the number of output currents of mirror is used to bias several other transistor circuits.

*** A current mirror with base-current compensation might be the solution to the problem.

↳ The second is the output resistance of the current source, which was found to be equal to the BJT r_0 and thus limited to the order of $100 k\Omega$.

** The need to increase the output resistance of current source can be seen if we recall that common mode gain of the differential amplifier is directly determined by the output resistance of its biasing current source.

*** A current mirror with emitter degeneration resistance R_E, will increase the output resistance of this (widlar current source) circuit.

Improved current-mirror circuit w/ base-current compensation:

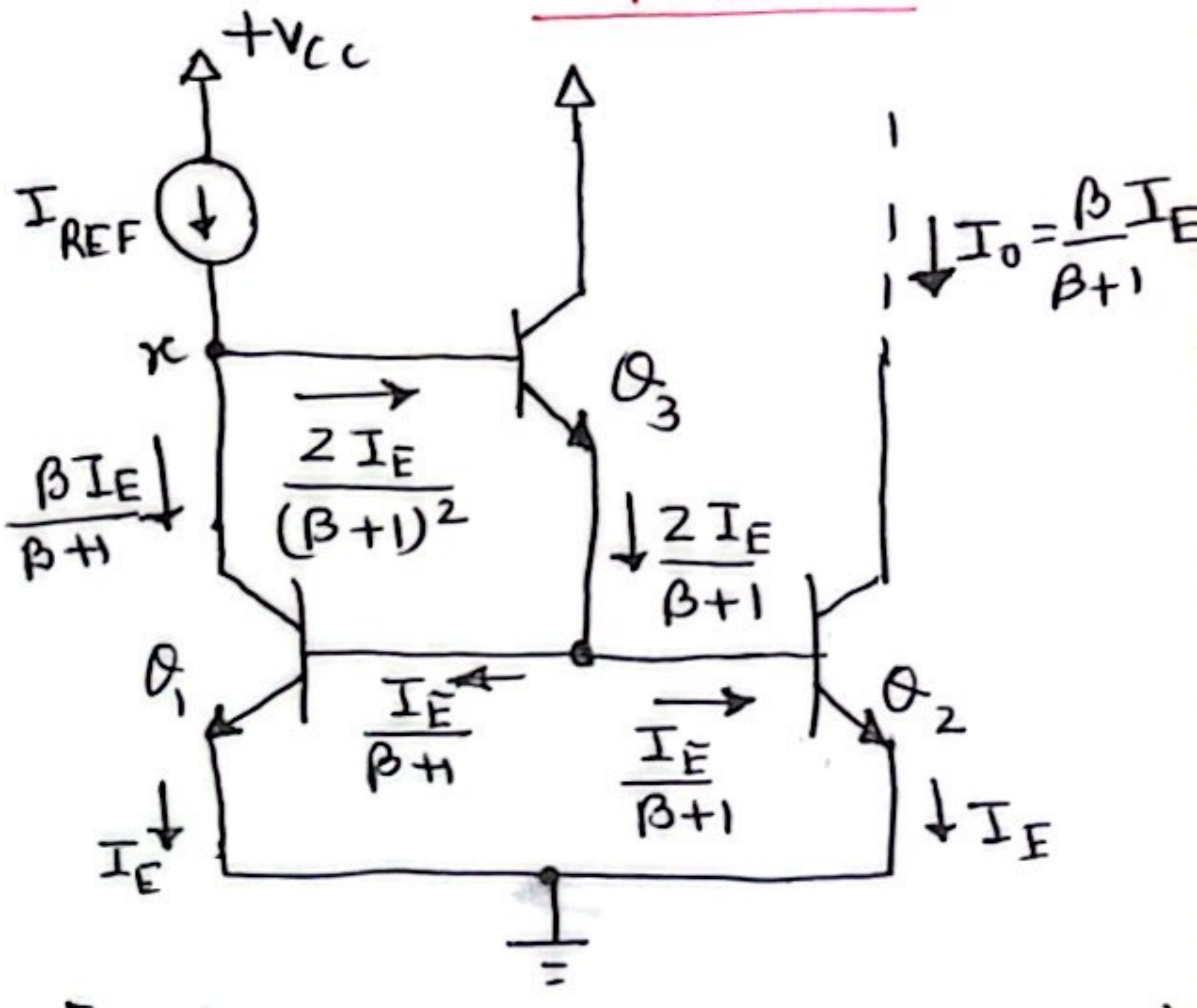

Fig. A current mirror with base-current compensation.

Above circuit includes a transistor Q_3 whose emitter supplies the base currents of Q_1 & Q_2. Almost all the base biasing current comes through the collector of Q_3. The sum of the base current divided by $(\beta+1)$, as base biasing current for Q_3 comes through reference current I_{REF}, which is much smaller current compare to basic current mirror circuit.

Here, Q_1, Q_2 and Q_3 are matched pair.

The reference current at node x can be written as

$$\boxed{I_{REF} = \frac{\beta}{\beta+1} I_E + \frac{2}{(\beta+1)^2} I_E} \quad -(1)$$

The output current is

$$\boxed{I_0 = \frac{\beta}{\beta+1} I_E} \quad ---(2)$$

The current gain of this mirror is

$$\frac{I_0}{I_{REF}} = \frac{1}{1+2/(\beta^2+\beta)} \boxed{\simeq \frac{1}{1+2/\beta^2}}$$

The above equation shows that the error due to finite β has been reduced from $2/\beta$ to $2/\beta^2$, a tremendous improvement.

To use the circuit as a current source, a resistance R_{ref} is connected between node x and the positive supply V_{CC}, then

$$\boxed{I_{REF} = \frac{V_{CC} - V_{BE_1} - V_{BE_3}}{R_{ref}}}$$

The Widlar Current Source:

Widlar current source differs from basic current mirror circuit in an important way: A resistor R_E is included in the emitter lead of Q_2.

Recall:

The collector current of BJT can be written as

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

or

$$\frac{I_C}{I_S} = e^{\frac{V_{BE}}{V_T}}$$

$$\boxed{\therefore \; V_{BE} = V_T \ln \frac{I_C}{I_S}}$$

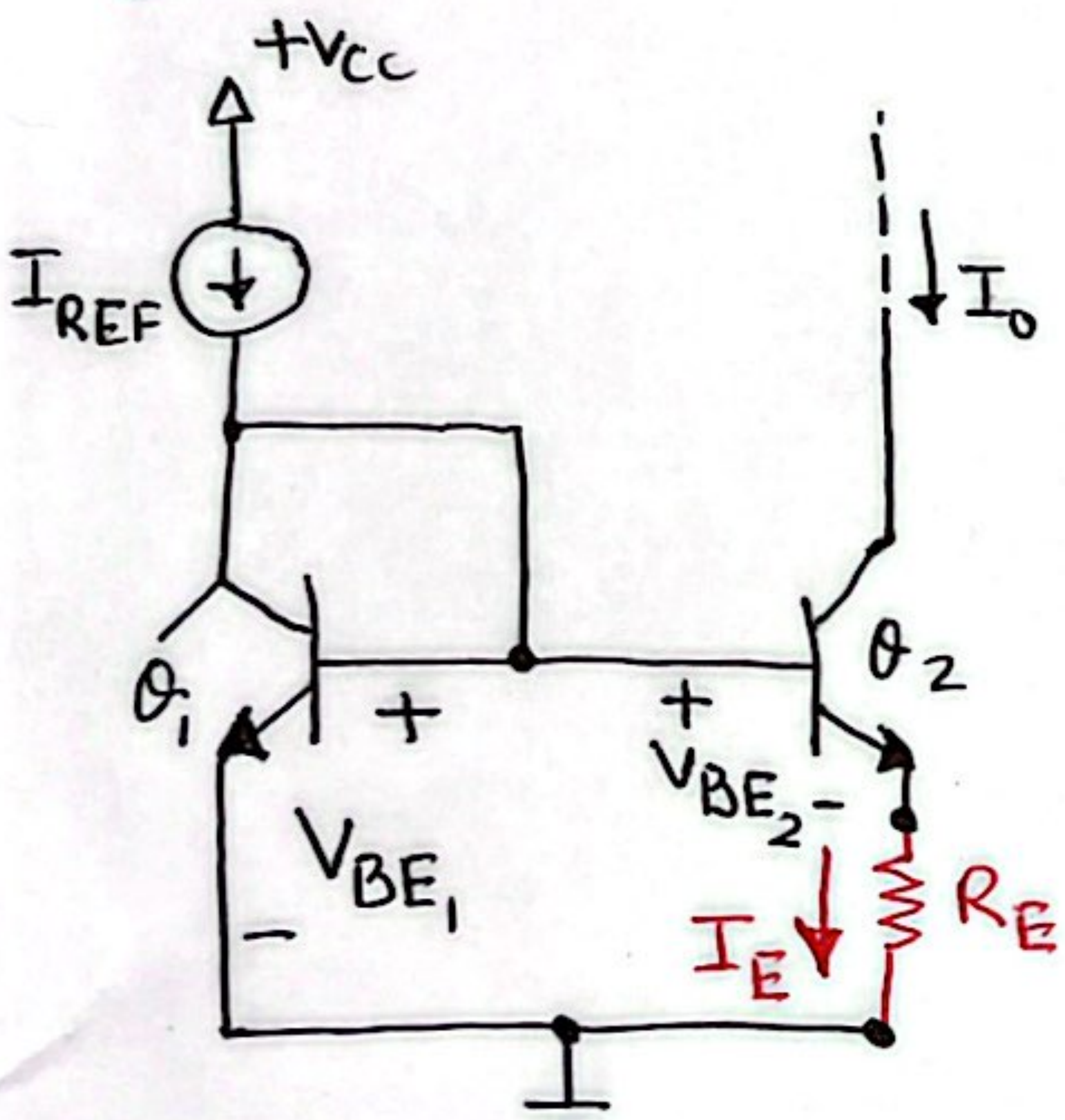

Fig. The widlar current Source

Neglecting base current,

$$\boxed{V_{BE_1} = V_T \ln \frac{I_{REF}}{I_S}} \quad \text{---} (1)$$

$$\boxed{V_{BE_2} = V_T \ln \frac{I_0}{I_S}} \quad \text{---} (2)$$

Assuming Q_1 & Q_2 are matched transistors, from eq^n (1) & (2)

$$\boxed{(V_{BE_1} - V_{BE_2}) = V_T \ln \left(\frac{I_{REF}}{I_0}\right)} \text{---} (3)$$

From circuit, we have

$$\boxed{V_{BE_1} = V_{BE_2} + I_0 R_E} \quad \text{---} (4)$$

Thus, from eq^n (3) & (4),

$$\boxed{I_0 R_E = V_T \ln \left(\frac{I_{REF}}{I_0}\right)}$$

* The transistor Q_1 is forward biased by V_{BE_1} while Q_2 is forward biased by V_{BE_2} only. Thus transistor Q_1 is more forward biased than Q_2. So the collector current of Q_2 (I_0) is much smaller than the collector current of Q_1 as I_{ref}.

** Therefore widlar current source is used to have much smaller output current (I_0) compare to reference current I_{ref}. (i.e. I_0 could be in μA range though I_{ref} is in mA range.)

*** The resistance R_{REF} and R_E could be of smaller value.

Therefore widlar current allows generation of a small constant current using relatively small resistors, which results in considerable savings in chip area.

Another important characteristic of widlar current source is that its output resistance is HIGH due to the presence of emitter resistance R_E

To Determine the output resistance of Widlar current Source:

To determine the output resistance of Q_2, the BJT is replaced with its Low-frequency hybrid-π model and apply a test voltage v_x to the collector of Q_2, as shown in fig.

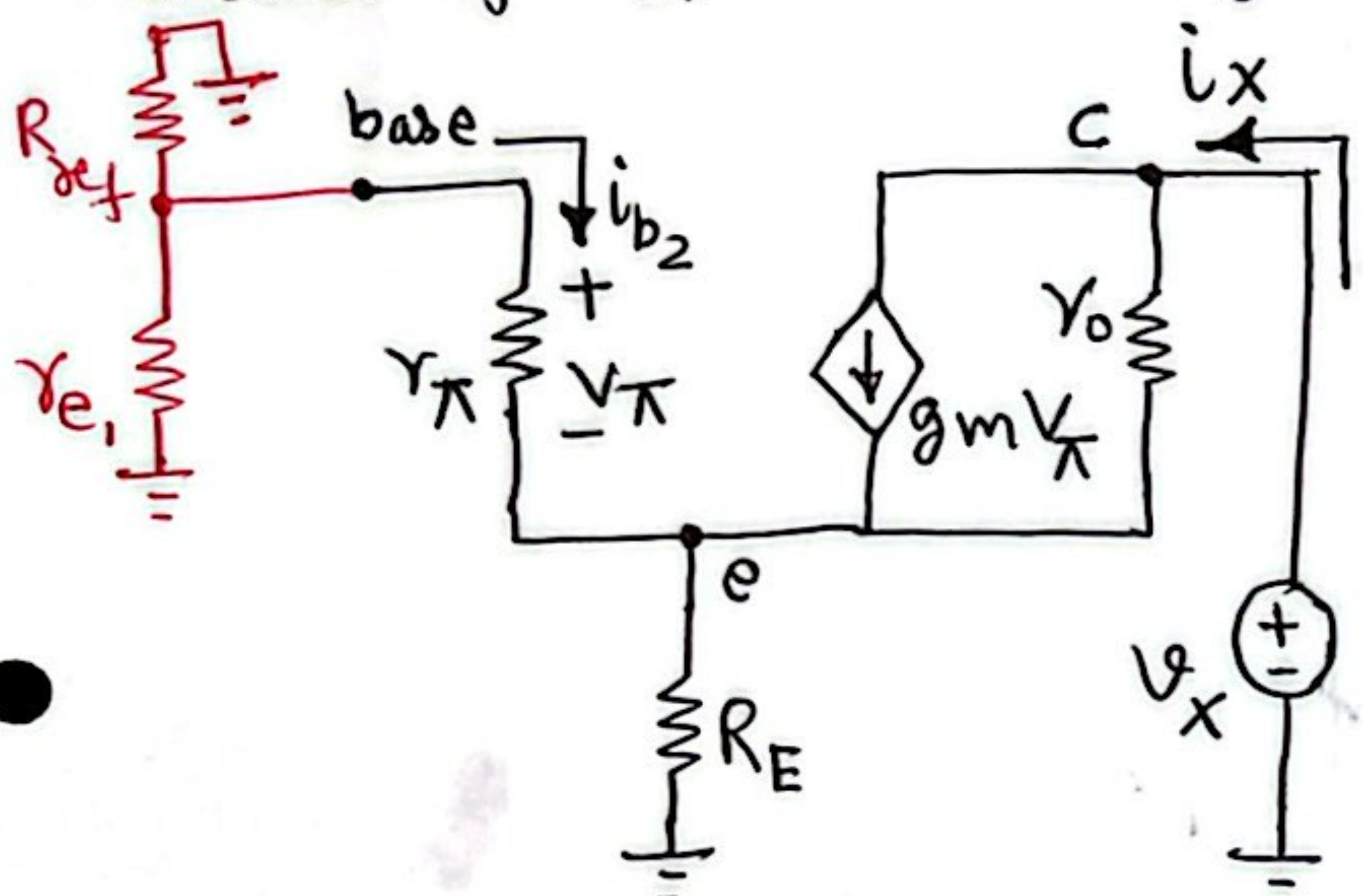

Fig. Small Signal Equivalent Circuit of Widlar Current Source.

* r_{e_1} is Small incremental resistance of diode connected transistor Q_1.
To simplify matters assume that this resistance is small enough to place the base of Q_2 at signal ground.

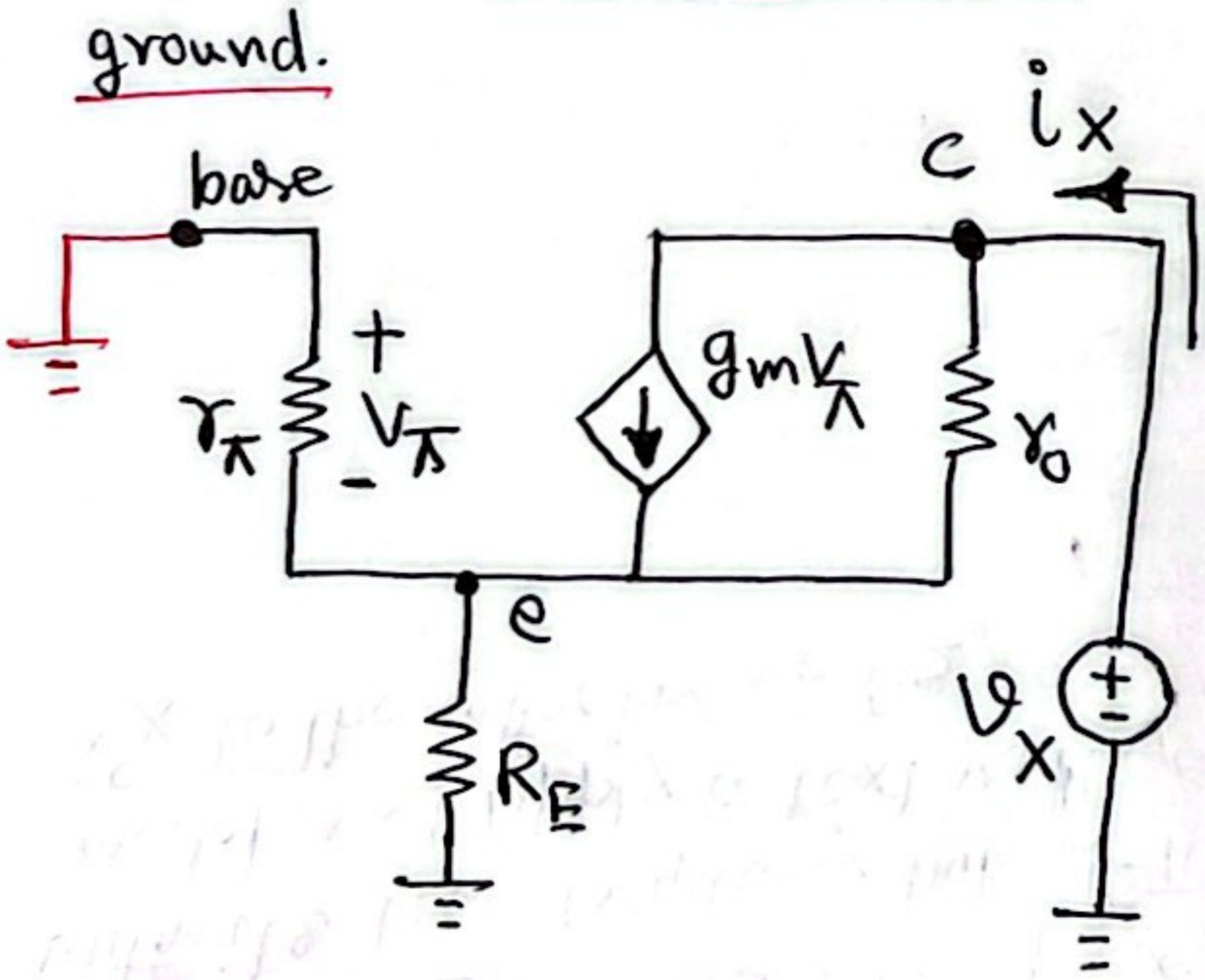

Fig Re-writting the circuit with base of Q_2 Connected to Signal ground.

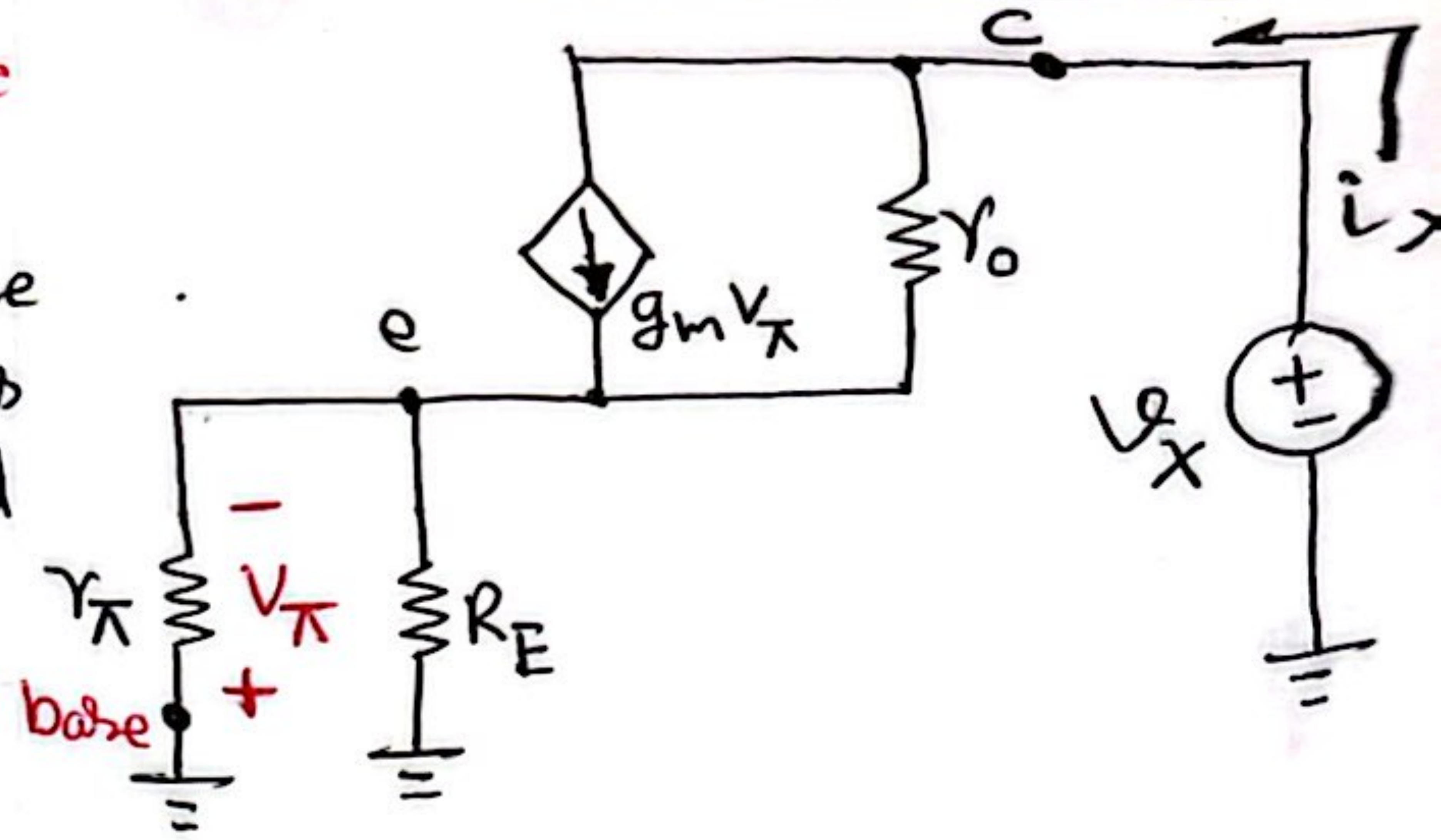

Fig Rewritting the circuit

Final Circuit can be written as

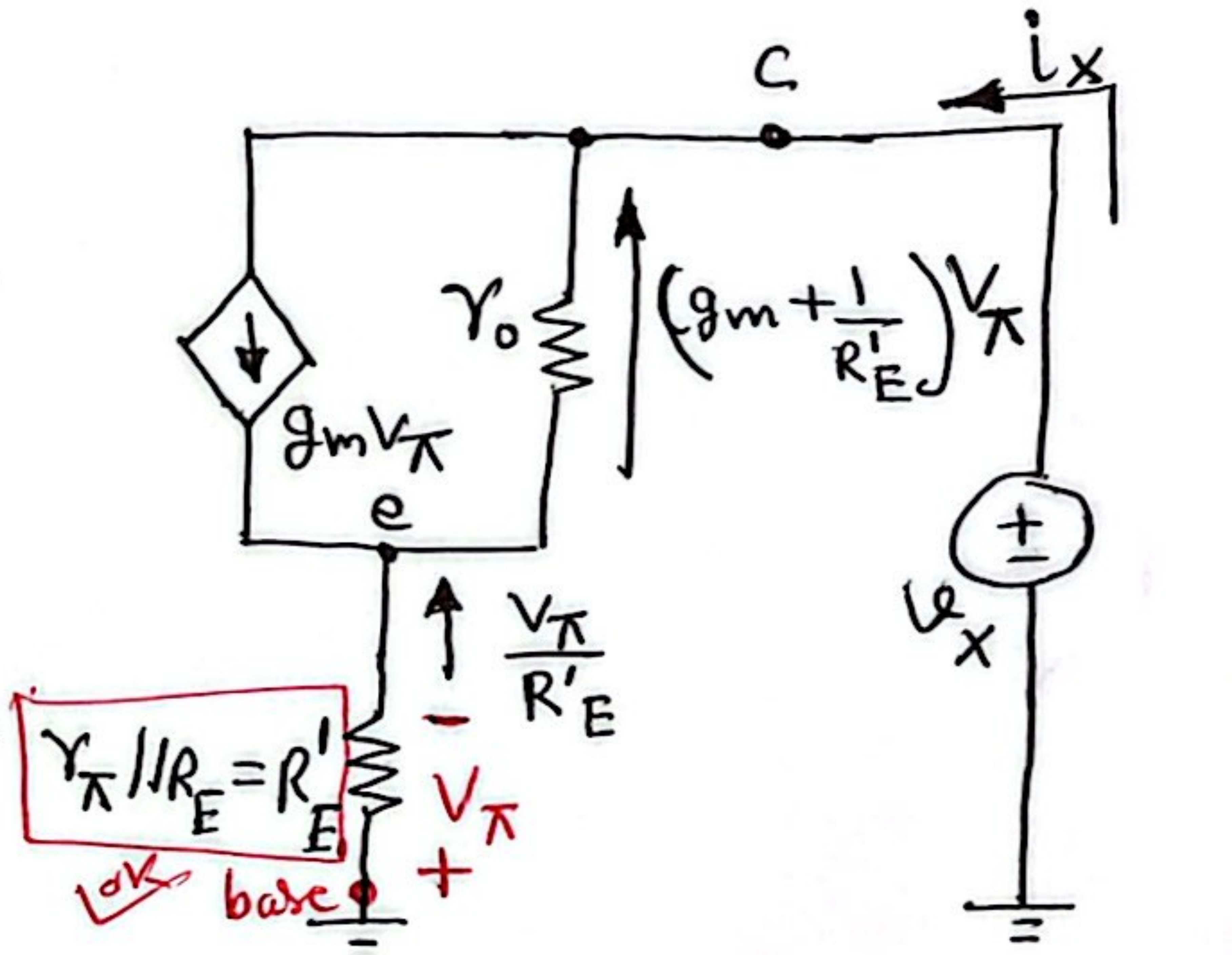

Fig. Determination of output resistance of Widlar current Source.

A Loop equation yields

$$\boxed{v_x = -\cancel{V_\pi} - \left(g_m+\frac{1}{R'_E}\right)V_\pi r_o} \quad \text{--- (1)}$$

and node equation at C provides

$$\boxed{i_x = g_m V_\pi - \left(g_m+\frac{1}{R'_E}\right)V_\pi} \quad \text{--- (2)}$$

Dividing eq^n (1) by eq^n (2) gives output resistance

$$R_o \equiv \frac{v_x}{i_x}$$

or $R_o = \dfrac{1+\left(g_m + \frac{1}{R'_E}\right) r_o}{1/R'_E}$

Which can be rearranged as

$$R_o = R'_E + (1+g_m R'_E) r_o$$

$$\simeq (1+g_m R'_E) r_o$$

Thus the output resistance is increased by the factor $1+g_m R'_E$ $= 1+g_m(R_E \| r_\pi)$.